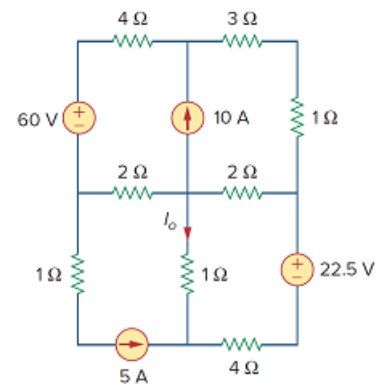


Problem

Apply mesh analysis to the circuit in Fig. 3.85 and obtain I_o .

Figure 3.85:



Step-by-step solution

Step 1 of 6

Refer to the circuit diagram in Figure 3.85 in the textbook.

The circuit is labeled and loop 3 and 4 form a super mesh as shown in Figure 2

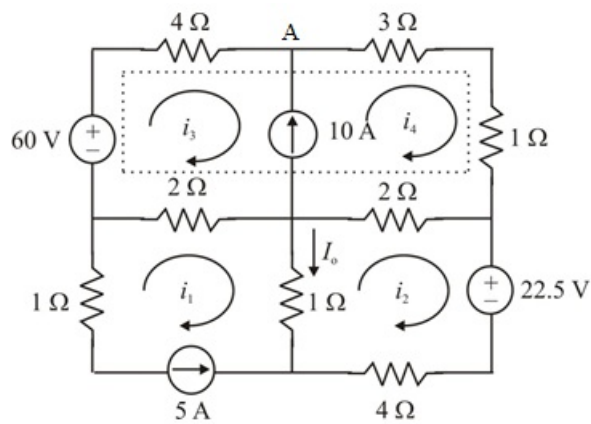


Figure 2

Step 2 of 6

The current i_1 is,

$$i_1 = -5 \text{ A}$$

Therefore $i_1 = -5 \text{ A}$ (1)

Apply Kirchhoff's voltage law around loop 2.

$$1(i_2 - i_1) + 2(i_2 - i_4) + 22.5 + 4i_2 = 0$$

$$i_2 - i_1 + 2i_2 - 2i_4 + 22.5 + 4i_2 = 0$$

$$-i_1 + 7i_2 - 2i_4 = -22.5$$

But $i_1 = -5 \text{ A}$

$$-(-5) + 7i_2 - 2i_4 = -22.5$$

$$7i_2 - 2i_4 = -27.5$$

Therefore $7i_2 - 2i_4 = -27.5$ (2)

Step 3 of 6

The loop 3 and 4 form a super loop.

Apply Kirchhoff's voltage law to super mesh.

$$-60 + 4i_3 + 3i_4 + i_4 + 2(i_4 - i_2) + 2(i_3 - i_1) = 0$$

$$-60 + 4i_3 + 3i_4 + i_4 + 2i_4 - 2i_2 + 2i_3 - 2i_1 = 0$$

$$-2i_1 - 2i_2 + 6i_3 + 6i_4 = 60$$

$$-i_1 - i_2 + 3i_3 + 3i_4 = 30$$

Therefore $-i_1 - i_2 + 3i_3 + 3i_4 = 30$ (3)

Apply Kirchhoff's current law at node A.

$$i_4 - i_3 = 10 \text{ A}$$

$$i_3 = i_4 - 10$$

Therefore $i_3 = i_4 - 10$ (4)

Step 4 of 6

Recall equation (3).

$$-i_1 - i_2 + 3i_3 + 3i_4 = 30$$

Substitute -5 for i_1 and $i_4 - 10$ for i_3 in the equation.

$$-(-5) - i_2 + 3(i_4 - 10) + 3i_4 = 30$$

$$5 - i_2 + 3i_4 - 30 + 3i_4 = 30$$

$$5 - i_2 + 6i_4 = 60$$

$$-i_2 + 6i_4 = 55$$

Therefore $-i_2 + 6i_4 = 55$ (5)

Recall equation (2).

$$7i_2 - 2i_4 = -27.5$$

Substitute $i_3 + 10$ for i_4 .

$$7i_2 - 2(10 + i_3) = -27.5$$

$$7i_2 - 20 - 2i_3 = -27.5$$

$$7i_2 - 2i_3 = -7.5$$

Therefore $7i_2 - 2i_3 = -7.5$ (6)

Step 5 of 6

Multiply equation (5) by 7 and adding equation (6) to it.

$$\begin{array}{r} 7i_2 - 2i_3 = -7.5 \\ -7i_2 + 42i_4 = 385 \\ \hline -2i_3 + 42i_4 = 377.5 \end{array}$$

Multiply equation (4) by 2 and add to the equation.

$$\begin{array}{r} 2i_3 - 2i_4 = -20 \\ -2i_3 + 42i_4 = 377.5 \\ \hline 40i_4 = 357.5 \\ i_4 = 8.93 \text{ A} \end{array}$$

Step 6 of 6

Now, find current i_3 .

$$i_3 = i_4 - 10$$

Substitute 8.93 A for i_4 in the equation.

$$\begin{aligned} i_3 &= 8.93 - 10 \\ &= -1.063 \text{ A} \end{aligned}$$

From equation (6), find the current i_2 .

Recall equation (6)

$$7i_2 - 2i_3 = -7.5$$

Substitute -1.063 A for i_3 in the equation.

$$\begin{aligned} 7i_2 - 2(-1.063) &= -7.5 \\ 7i_2 &= -7.5 - 2.125 \\ i_2 &= \frac{-9.625}{7} \\ &= -1.375 \text{ A} \end{aligned}$$

The current I_o can be calculated as:

$$\begin{aligned} I_o &= i_1 - i_2 \\ &= -5 - (-1.375) \\ &= -3.625 \text{ A} \end{aligned}$$

Therefore the current I_o is -3.625 A.